

A Hybrid Approach to Process Mining: Finding Immediate Successors of a Process by Using From-To Chart

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Abstract—Process mining is a branch of data mining that aims to discover process model from the event logs. In this study, we propose a hybrid approach to process mining in such a way that, “from-to chart” is used as the front-end to monitor the transitions among activities of a realistic event log. Another novelty of this study is developed evaluation metrics, which are used for finding immediate successors in order to convert these raw relations into dependency/frequency graph.

Index Terms—From-to Chart, Process Discovery, Immediate Successor, Process Mining, Event Logs

I. INTRODUCTION

Even though contemporary information systems are intensively utilized in enterprises, their actual impact in automating complex business processes is still constrained by the difficulties encountered in the design phase [8]. Actually crucial problems are resulting from the discrepancy between process design and process enactment [11, 14]. Process mining aims to improve the process modeling task by discovering patterns from the event logs for either improving an existing process model or discovering the process model without the guidance of a priori model.

In the scope of this study, we aim to develop a from-to chart based process mining methodology for finding immediate successors of activities in a process. *From-to chart* is an analytical tool, which is basically used in monitoring material handling routes between operations, machines, departments or work centers on the production floor [4]. Proposed methodology inherits this tool from industrial engineering domain and implements it in a distinct field, i.e. process discovery, as the basic bookkeeping material in monitoring transitions among activities occurred in process instances and figuring out if there exists any specific order of the occurrences for representing in process model.

Main cutting edge characteristic of the proposed approach is the implementation of from-to chart, as the front-end to monitor the transitions among activities in process instances, which are embedded in event logs, and convert the raw

successive relations into an activity sequence, which is optimum due to moment minimization.

This paper is organized as follows: Section 2 includes the related work. Section 3 describes from-to chart. Section 4 presents the proposed approach. In Section 5, the application of the presented approach on a real case is described. Finally, Section 6 includes the concluding remarks.

II. RELATED WORK

In [2], Aalst et al. highlights the process mining usage in analysis and improvement of process flexibility. It proposes the enrichment of obtained event logs with semantic information. Consequently process mining techniques can capture the reactive and stimulus entities for the change in event logs.

Likewise [2], [9] aims to handle both case and control-flow perspectives in an integrated fashion. While production workflow is strongly process-oriented, the case handling paradigm basically concentrates on the case itself, which is the fundamental object to be produced.

The idea of applying process mining in the context of workflow management was first introduced in [3]. In this study, two issues are handled. The first issue is to discover a workflow graph generating activities appearing in a given workflow log. The second issue is to find the definitions of the relation conditions.

Cook and Wolf have investigated similar issues in the context of software engineering process. In [5], they describe three methodologies for process discovery ranging from the purely algorithmic to purely statistical: one using neural networks named RNET, one using a purely algorithmic approach named KTAIL and one Markovian approach named MARKOV.

In [13], Weijters and Aalst propose a rediscovery technique to deal with noise and to validate workflow processes by uncovering and measuring the discrepancies between the prescriptive models (e.g. reference models) and actual process executions. Detecting concurrency appears as one of the fundamental concerns. Therefore AND/OR connectors are aimed to be explicitly distinguished in the process model. To accomplish this goal, WorkFlow nets, which are a subset of Petri nets, are combined with techniques from machine learning. Moreover, local and global metrics are proposed to find explicit representations for a broad range of process models. These metrics are revised in [11]. In [1], the proposed method, named “alpha algorithm”, seeks to

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rediscover sound Workflow nets. Given such an algorithm, it is aimed to find the class of workflow nets which can be rediscovered. As a way of representation, it generates concrete Petri nets for a broad range of process models rather than a set of dependency relation between events like in [13].

As a hybrid methodology in process mining, Gomez et al. introduce “Application Usage Mining” concept in [7]. Application usage mining is explained as the examination of the user’s behavior in the business application systems (e.g. ERP) by applying the basic approaches, notions and methods inherited from web usage mining.

III. FROM-TO CHART

The basic from-to chart is a square, symmetric matrix for summarizing material handling between related operations, machines, departments or work centers on the production floor [4] with high volume production rate [6]. The sequence of operations is written down the left-hand side of the form and across the top. While the vertical sequence of activities is the “from” side of the matrix, the horizontal sequence of activities is the “to” matrix [12]. This analytical technique is useful for several purposes including designing relative locations of operations, demonstrating material flow patterns and showing degree of self-sufficiency of each operation [4].

Basic data for entry into from-to chart are prepared by tabulating the flow paths of each concentrated part, product or material such that; for each move of related entity from operation i to operation j , current score at the $(i, j)^{th}$ element of matrix is incremented by one. Thus, accumulated scores in each element represent the total number of moves from and to the underlying operation.

Constructed from-to chart has to be analyzed for better arrangements of operations to reduce handling, costs, distances, production control problems, etc. [4]. Major use-cases occurred at from-to chart are as follows:

- i. All entries below the diagonal indicate *back-tracking*, i.e., backwards from the order indicated by the numbers representing the operations.
- ii. All entries in the upper right or far right indicate *skipping* past several adjacent operations to get to their next operation.
- iii. Items moving from one operation to an adjacent operation result in the marks falling in the elements along and just above the diagonal. This represents *straight-line* (direct) flow.

Major use cases are visualized at from-to chart in Figure 1.

TO \ FROM	Rough Stores	Mill	Lathe	Drill	Bore	Grind	Press	Hone	Saw	Final Inspection
Rough Stores		2	8			1	4		2	
Mill			1	2			1			1
Lathe			2	4			1		1	3
Drill					1		2	1		5
Bore				1						
Grind					1					1
Press				2						6
Hone										1
Saw		2				1				
Final Inspection										

■ Back-tracking
 ■ Skipping
 ■ Straight-line (Direct)

Figure 1. Major Use Cases on the From-to Chart

Intuitively it is seen that the best layout can be devised by rearranging the columns and rows to put the elements with larger scores just above the diagonal and fewer ones below the

line. According to [6], from-to chart is a descriptive material to reduce a large volume data into a workable formation such that, the construction of a from-to chart does not result directly in the solution of a layout problem. On the other hand, a more quantitative approach to minimize material handling is obtained by taking “moments” of the accumulated score at each element around the diagonal (i.e. support point) and aiming for the lowest moment total at from-to chart as stated in [4]. The number of elements away from the diagonal is used as the distance from the diagonal, i.e. moment arm. Objective function to minimize the total moment of from-to chart is formulated as follows:

$$\text{Min } Z = \sum_{i=1}^N \sum_{j=1}^N f_{ij} \times |j - i| \times p$$

In this formula, f_{ij} indicates total move from operation i to operation j , $|j - i|$ is the distance from operation i to operation j , and p is the back-tracking penalty point assigned to each entry below the diagonal. Back-tracking penalty point is doubled to enforce the model towards a straight line arrangement [12]. Therefore, p is set as follows:

$$p = \begin{cases} 1 & \text{when } j \geq i \\ 2 & \text{when } i > j \end{cases}$$

IV. FINDING IMMEDIATE SUCCESSORS BY USING FROM-TO CHART

A. Constructing From-To Chart from Event Logs

The starting point of the proposed methodology is the creation of a so-called FROMTOCHART table by retrieving the activity titles from the event logs and populating FROMTOCHART table. For populating the table, event logs are arranged by process instances and then ordered by timestamp in ascending order. Following this, predecessor and successor activities are parsed for each transition in activity streams. The current tally mark of (predecessor, successor)th element in FROMTOCHART table is incremented by one. As a result, all transitions among activities in process instances are marked to FROMTOCHART table.

B. Evaluating the Tally Marks

In traditional from-to chart implementation, total score of each element is directly taken into consideration in rearrangement of the matrix. However, by pruning down the “weak” scores prior to rearrangement, it is aimed to eliminate their effect on the optimum activity sequence in the proposed methodology. Developed evaluation metrics are similar to the original formations in association rule mining [10] except for the predecessor and successor notations. There are three evaluation metrics used in the proposed methodology:

a. Confidence for From-To Chart (Confidence FTC): It is the ratio of transitions, which are from predecessor A to successor B, to the total number of transitions which are

initiated by activity A (i.e. row total of activity A in FROMTOCHART table).

$$\text{Confidence FTC} = \frac{|A > B|}{|A > *|}$$

In this formula, $|A > B|$ is the total number of transitions from activity A to activity B , and $|A > *|$ is the total number of transitions initiated by activity A (row total of activity A in FROMTOCHART table). Confidence FTC takes into account the magnitude of $|A > B|$ and $|A > *|$. In this aspect, it is analogous to local metric (LM) stated in [11].

b. Support for From-To Chart (Support FTC): The previous metric is for expressing the tendency of succession by comparing the magnitude of $|A > B|$ and $|A > *|$ at local level. However, for determining the likelihood of succession between two activities A and B , *support for from-to chart (Support FTC)* is indeed a global metric because it takes into account the overall process instances. *Support for from-to chart* is the ratio of transitions, which are from predecessor A to successor B , to the total number of process instances dedicated to training dataset (i.e. $\#L$).

$$\text{Support FTC} = \frac{|A > B|}{\#L}$$

In this formula, $|A > B|$ is the total number of transitions from activity A to activity B and $\#L$ is the total number of process instances in training dataset. This metric is similar to global metric (GM), given in [11].

The level of robustness in discovered process model can be easily controlled by the end-user or domain expert through the probability threshold parameters. These parameters can also be used to manage the complexity of the discovered model from large amounts of data; eliminating low-probability sequences, even if they are not due to the noise factor [5].

c. Modified Lift: Support and confidence measures are practical barriers, which are “supervised” by end-user. Thus lowered support FTC and confidence FTC values may encourage this methodology by representing all patterns exhibited in training dataset and they may turn into insufficient evaluation metrics at filtering out uninteresting rules. To tackle this weakness, a correlation measure named modified lift can be used to augment the support-confidence framework.

Modified lift is a simple correlation measure that is adapted from conventional lift measure [10] and it is defined as follows.

$$\text{Modified lift } (A, B) = \frac{|A > B| \times \#G}{|A > *| \times |* > B|}$$

In this formula, $|A > B|$ is the total number of transitions from activity A to activity B , $|A > *|$ is the total number of transitions initiated by activity A (row total of activity A in FROMTOCHART table), $|* > B|$ is total number of transitions attained to activity B (column total of activity B in

FROMTOCHART table), and $\#G$ is gross total tally mark in FROMTOCHART table.

Major modification performed on original lift measure is the evaluation procedure of tally marks with respect to calculated lift value such that;

- If the resulting value of modified lift is greater than 1, then activities A and B are positively correlated, meaning that the occurrence of activity A triggers the occurrence of activity B . Thus tally mark of element (A, B) does not change.
- If the resulting value of modified lift is equal to 1, then activities A and B are independent and there is not any correlation between these activities. Thus tally mark of element (A, B) is reset to zero.
- If the resulting value of modified lift is less than 1, then activities A and B are negatively correlated, meaning that the occurrence of activity A discourages the occurrence of activity B . Thus tally mark of element (A, B) is multiplied by minus 1.

This “minus one” factor is inherited from “the Big M method” in linear programming (LP). When the Big M method is applied, the major argument is to determine “how large M should be”. Because extremely large coefficients may dominate objective function and deviate the optimum. In this aspect, multiplying tally mark by minus 1 can be evaluated as a “realistic” value to instruct the rearrangement operation that activity A and B are negatively correlated without affecting optimum activity sequence. Consequently rearrangement operation tends to separate these activity pairs due to decrement in moment calculation.

C. Rearranging From-To Chart

This operation is the *engine component* of proposed methodology, which aims to find out the optimum activity sequence with the minimum overall moment value in FROMTOCHART table. Intuitively, the best sequence is obtained by positioning the elements with higher tally marks along and above the diagonal. Hence straight-line transitions between the activities with higher inter-traffic are concentrated.

The initial step in rearrangement of FROMTOCHART table is to generate activity sequences. This is performed by Permutation object, which creates activity sequences from the seed set of activity types observed in event log. Each generated activity sequence is checked whether the first element matches the initiator activity, which is anticipated by domain expert. If it does not match, this sequence is discarded. Otherwise, records in FROMTOCHART table are retrieved according to the order emphasized at the generated activity sequence. Indeed this order is crucial to calculate the “moment arm” accurately.

In the following step, each record’s moment value is calculated locally. The most significant notion in this step is the diagonal (e.g. support point). Tally marks at the underlying row are multiplied by relative distance of concentrated column to the diagonal. In the case of back-tracking (concentrated column < diagonal), calculated moment value is multiplied by back-tracking penalty point. By this way, back-tracking use cases are discouraged in the optimum activity sequence.

After all records are traversed, calculated total moment value is compared to local optimum total moment value. The smaller value becomes the new local optimum total moment value. At the end of the process, local optimum total moment value and local optimum activity sequence are set as global optimum value and sequence.

D. Extracting Relations

Although global optimum activity sequence is the backbone of discovered process model, it is limited to sequential behavior. In this aspect, this operation aims to extract one-to-one and one-to-many relations among activities by concentrating on the evaluated tally marks in FROMTOCHART table.

Evaluated tally marks are the essential stick yard to determine whether the underlying transition is worth to be visualized in process model such that;

- i. Tally mark at $(i,j)^{th}$ element, which is greater than zero, highlights that the transition represented by this element is significant with respect to evaluation metric(s). Therefore successor of the transition (i.e. activity j) is added to successive activity list of activity i .
- ii. Negative or zero-valued tally mark indicates that the underlying transition is challenged by evaluation metric(s). Therefore this element is skipped in constructing process model operation.

According to the rule stated above, three types of relations are defined on the process model:

a. Immediate Succession: Let S be an optimum activity sequence over T , a range of activities, i.e. $S \in T^*$ (i.e. a permutation of T) and $A, B \in T$. Then B immediately succeeds A , if and only if position of A (p_A) and position of B (p_B) at S are two successive integers such that; $p_B = p_A + 1$, where p_A and $p_B \in [1, |S|]$. This implication is similar to direct succession definition stated in [11].

b. Succession: Let S be an optimum activity sequence over T , a range of activities, i.e. $S \in T^*$ (i.e. a permutation of T) and $A, B \in T$. Then B succeeds A , if and only if position of A (p_A) and position of B (p_B) at S are two integers such that; $p_B > p_A$, where p_A and $p_B \in [1, |S|]$. This implication is similar to succession definition stated in [11].

c. Back-tracking: Let S be an optimum activity sequence over T , a range of activities, i.e. $S \in T^*$ (i.e. a permutation of T) and $A, B \in T$. Then B backtracks A if and only if position of A (p_A) and position of B (p_B) at S are two integers such that; $p_B < p_A$, where p_A and $p_B \in [1, |S|]$.

V. EXTRACTING IMMEDIATE SUCCESSORS: TRAVEL MANAGEMENT BUSINESS PROCESS CASE

Travel management business process is built as a sub-component of the finance module (i.e. SAP FI/CO) on an ERP-system at 64 provincial offices of a public institution. Reference model consists of seven activity types, which are “create travel request”, “change travel request”, “display travel request”, “advance payment”, “travel request confirmation”, “disbursement entry” and “deduction”. Table 1 shows the initial state of FROMTOCHART table for travel management process logs. Note that create activity is denoted as the initiator activity.

Table 1. Initial State of FROMTOCHART Table

	Create	Advance	Confirm	Change	Display	Disbursement	Deduction	Row Total
Create	0	291	102	44	38	0	0	475
Advance	0	0	285	13	38	0	2	338
Confirm	0	0	1	4	21	312	137	475
Change	0	27	29	0	5	0	0	61
Display	0	20	60	0	0	13	27	120
Disbursement	0	0	0	0	17	0	307	324
Deduction	0	0	0	0	0	0	0	0

After FROMTOCHART table is created and parsed transitions in activity streams are populated to this table, tally marks are evaluated with respect to support and confidence thresholds (support $FTC=0.15$ and confidence $FTC=0.15$). Theoretically calculated support threshold acts as a upper-bound for evaluating tally marks in FROMTOCHART table, since total number of process instances in training dataset ($\#L$) is the maximum row total for unrepeated activity types. Therefore resetting is mostly performed due to calculated support threshold. Table 2 shows the final state of FROMTOCHART table.

Table 2. FROMTOCHART Table after Tally Mark Evaluation

	Create	Advance	Confirm	Change	Display	Disbursement	Deduction	Calculated Support Threshold	Calculated Confidence Threshold
Create	0	291	102	0	0	0	0	71.25	71.25
Advance	0	0	285	0	0	0	0	71.25	50.70
Confirm	0	0	0	0	0	312	137	71.25	71.25
Change	0	0	0	0	0	0	0	71.25	9.15
Display	0	0	0	0	0	0	0	71.25	18.00
Disbursement	0	0	0	0	0	0	307	71.25	48.60
Deduction	0	0	0	0	0	0	0	71.25	0.00

Calculated Support Threshold = $0.15 * 475$ (number of process instances in training dataset)
Calculated Confidence Threshold = $0.15 * \text{row total of underlying sending activity}$

After tally marks are evaluated, total moment value for each activity sequence starting with create activity is calculated. For instance, the first generated activity sequence is {create, advance, confirm, change, display, disbursement, deduction}. Total moment for this activity sequence is calculated as follows:

$$\begin{aligned}
 \text{Moment}_{\text{create}} &= 1 \times 291 + 2 \times 102 & \text{Moment}_{\text{advance}} &= 1 \times 285 \\
 \text{Moment}_{\text{confirm}} &= 3 \times 312 + 4 \times 137 & \text{Moment}_{\text{disbursement}} &= 1 \times 307 \\
 \text{Total moment} &= 2571
 \end{aligned}$$

Since the activities change, and display increment the moment arms of activities disbursement and deduction at the row belonging to confirm activity, these activities have to be shifted to the end of activity sequence. Hence global optimum activity sequence is obtained as {create,

advance, confirm, disbursement, deduction, change,
display}.

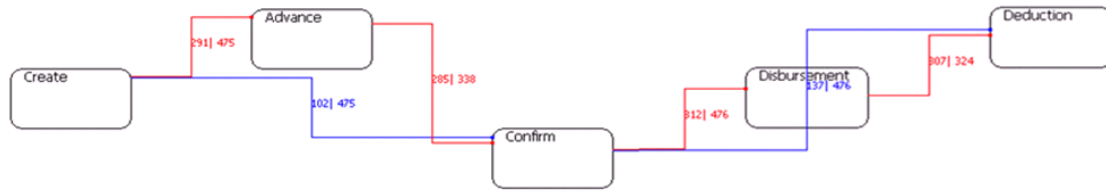


Figure 2. Discovered Relations in Dependency/Frequency Graph Form

After rearrangement of FROMTOCHART table, relations are extracted according to positive tally marks remarked in FROMTOCHART table. Consequently, dependency/frequency graph, which is a kind of finite state machine (FSM) that visually represents derived relations and their dependency scores, is constructed as shown in Figure 2.

For this data set, the ratio of immediate successors that are correctly found in reference model is 100%. Additionally, all extracted immediate successors have corresponding complements in testing dataset. Therefore we can conclude that proposed methodology has a satisfactory performance of determining immediate successors.

VI. CONCLUSION

In this work, we propose a hybrid approach for deriving relations among activities from event logs. From-to chart, which is inherited from industrial engineering domain, is a fundamental analytical tool used in monitoring material handling routes. The novelty of this methodology resides in the fact that; an analytical tool from industrial engineering is adapted to a new problem domain and is used as the baseline to monitor transitions among activities observed in business process. The activity sequence, which is rearranged in a straight-line form according to optimization on from-to chart, constitutes the backbone of discovered process model.

Developed evaluation metrics (i.e. support FTC, confidence FTC and modified lift) act as the major stick yard to control the level of robustness and complexity of the discovered process model from large amounts of data. While formulating these metrics, the original formations for evaluations in association rule mining are taken as the basis.

As the future work, the generated activity sequence may be further improved by analyzing from-to chart for discovering branchings and loops in the process model.

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